

Car Sim Perturbation Study — Noise & Resolution (subset)

1. Motivation & Context

- Replays the MNIST noise/downsampling analysis on the `car_sim` renders to understand how colour variants react when the input is corrupted.
- Focuses on embedding geometry and predictive diagnostics from CLIP ViT-B/32 under Monte Carlo Dropout (MCDO).
- Uses a deliberately tiny evaluation slice (three views per colour) to respect the request for “first, middle, last” coverage while keeping the run tractable.

2. Experimental Configuration

- **Data:** 21 RGB crops derived by taking the first, middle, and last frame from each colour folder under `data/car_sim`; copied into `data/car_sim_subset`.
- **Backbone:** `openai/clip-vit-base-patch32`, no adapters or extra dropout beyond CLIP defaults.
- **Sampling:** 32 stochastic passes per image, micro-batch size 4, temperature $\tau = 1.0$, deterministic seed 0, predictive head enabled.
- **Outputs:** Scenario folders plus `aggregated_metrics.csvjson` written to `runs/car_sim_perturbations_noise_res_subset/`. Derived tables live under `docs/report_assets/car_sim/`.
- **Command:**

```
PYTHONPATH=src \  
python -m uclip.cli.car_sim_perturbation_study \  
  --root data/car_sim_subset \  
  --out-root runs/car_sim_perturbations_noise_res_subset \  
  --passes 32 \  
  --microbatch 4 \  
  --limit 0 \  
  --noise-stds 0.0,0.05,0.1,0.2,0.35 \  
  --downsample-sizes 144,104,72
```

3. Perturbation Catalogue

- **Baseline:** Original crops processed by the CLIP image encoder.
- **Gaussian noise:** Additive $\sigma \in \{0.05, 0.10, 0.20, 0.35\}$ in $[0, 1]$, clipped back to valid intensities before CLIP preprocessing.
- **Spatial downsampling:** Short side resized to $\{144, 104, 72\}$ px with bilinear filtering, then upsampled back to the native resolution. All scenarios share prompts generated from folder names (e.g., “a photo of a Blue jeep”).

4. Metrics of Interest

- **Trace ($\text{Tr } \Sigma$)** — total variance across the 512-D embedding cloud; constant across scenarios here due to identical dropout instrumentation.
- **Log-determinant ($\log \det \Sigma$)** — covariance volume proxy (also numerically flat in this subset).
- **Off-diagonal mass** — L_1 sum of covariance off-diagonals, highlighting cross-dimensional coupling shifts.
- **Predictive diagnostics** — top-1 accuracy, entropy, mutual information, and confidence from the text head prompts. Baseline reference: trace = 5.12×10^{-4} , logdet = -7073.54, off-diagonal mass = 4.38×10^{-9} , accuracy = 0.619.

5. Results

5.1 Aggregate behaviour under noise

σ	Accuracy	Off-diag ($\times 1e-9$)	Δ off-diag (%)
0.05	0.714	4.143	-5.31
0.10	0.667	4.264	-2.56
0.20	0.667	4.073	-6.92
0.35	0.667	3.956	-9.59

- Mild noise ($\sigma = 0.05$) slightly *improves* accuracy (+9 percentage points vs. baseline) and lowers off-diagonal mass, suggesting the perturbation regularises colour cues.
- Heavier noise trims the confidence-oriented terms (confidence down from 0.150 to 0.146) while keeping accuracy flat; covariance volume remains unchanged within numerical precision.
- Tabular export: docs/report_assets/car_sim/noise_summary.csv.

5.2 Aggregate behaviour under downsampling

Short side	Accuracy	Off-diag ($\times 1e-9$)	Δ off-diag (%)
144	0.619	4.291	-1.93
104	0.619	4.295	-1.85
72	0.619	4.529	3.51

- Resolution degradation keeps accuracy anchored at the baseline 0.619, but off-diagonal mass first contracts (blur dampens cross-dimension coupling) before rebounding at the coarsest 72 px setting.
- Even with aggressive downsampling, trace and log-determinant stay numerically identical to baseline, reinforcing that off-diagonal movement carries the discriminative signal.
- Tabular export: docs/report_assets/car_sim/downsample_summary.csv.

5.3 Class-level sensitivity snapshot

Class	Baseline acc	$\sigma=0.35$ acc	72px acc
Jeep Blue	0.00	0.33	0.00
Jeep Green	1.00	1.00	1.00
Jeep Indigo	0.33	1.00	0.33
Jeep Orange	0.00	0.00	0.00
Jeep Red	1.00	1.00	1.00
Jeep Violet	1.00	0.67	1.00
Jeep Yellow	1.00	0.67	1.00

- Blue and Orange jeeps stay challenging regardless of perturbation, hinting that CLIP's text prompts struggle to disambiguate those hues with only three exemplars.
- Indigo benefits from stronger noise (3/3 correct) but falls back under heavy downsampling, suggesting texture cues matter more than shape.
- Confidence scores shrink most for the problematic colours (≈ -0.005 absolute under $\sigma = 0.35$), reinforcing the qualitative difficulty.

5.4 Takeaways

1. **Covariance volume is stable** — With only three samples per class, trace/logdet barely move; off-diagonal shifts carry the signal.
2. **Noise can regularise** — Light Gaussian noise boosts accuracy by smoothing high-frequency artefacts.
3. **Hard classes persist** — Blue and Orange variants remain failure cases, indicating prompt or data augmentation work is needed more than brute-force sampling.

4. **Subset limitations** — Results illustrate how the new CLI behaves, but the tiny dataset magnifies discrete effects (accuracy jumps of 0.333) and should not be overgeneralised.

6. Limitations & Next Steps

- **Sample count:** Three frames per colour exaggerate step-changes. Re-running with fuller coverage will stabilise statistics once the pipeline is trusted.
- **Prompt fidelity:** The template prompts may underspecify colour nuances; richer wording or per-class prompt tuning could improve the stubborn Blue/Orange cases.
- **Additional corruptions:** Extending to brightness, contrast, or occlusion would reveal whether the observed robustness carries beyond noise/blur.
- **Next actions:**
 1. Re-run the perturbation sweep on the full dataset (or larger stratified slice) now that the CLI path is validated.
 2. Prototype prompt variants (e.g., “a studio photo of a bright blue jeep”) and compare predictive metrics using the same subset for apples-to-apples measurement.